

One frequency: unlimited duration

$$x(t) = e^{i2\pi f_0 t} \longleftrightarrow X(f) = \delta(f - f_0).$$

- A single-frequency complex exponential extends indefinitely in time.
- A real sinusoid contains the two frequencies $+f_0$ and $-f_0$:

$$\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)].$$

- Both signals are completely delocalized in time.

An ideal limiting case

These signals do not have finite energy. Their time spread is unlimited; we do not directly apply finite-energy standard-deviation formulas.

Perfect localization requires all frequencies

$$\delta(t - t_0) \longleftrightarrow e^{-i2\pi ft_0}$$

- A Dirac impulse is concentrated at one instant.
- Its spectrum has equal magnitude at every frequency.
- Its phases are arranged to concentrate the signal at t_0 .

All frequencies alone are insufficient

Their relative amplitudes and phases determine the resulting signal. Unlimited bandwidth does not, by itself, imply zero duration.

The impulse is also an ideal limit, rather than a finite-energy signal.

Fourier uncertainty concerns the signal

For a nonzero finite-energy signal, use normalized energy distributions:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df.$$

$$\sigma_t^2 = \frac{1}{E} \int_{-\infty}^{\infty} (t - \bar{t})^2 |x(t)|^2 dt, \quad \sigma_f^2 = \frac{1}{E} \int_{-\infty}^{\infty} (f - \bar{f})^2 |X(f)|^2 df.$$

Here $\bar{t}$ and $\bar{f}$ are the energy centroids.

Localization principle (when both variances are finite)

$$\sigma_t \sigma_f \geq \frac{1}{4\pi} \quad (f \text{ in Hz}).$$

Concentration in one domain requires spread in the other.

This is a property of Fourier pairs, independent of sampling. The infinite extent of the basis functions provides intuition.

Sampling concerns the axes

For N samples with interval δt , the standard DFT uses

$$\boxed{T = N\delta t}, \quad \boxed{\delta f = \frac{1}{T}}, \quad \boxed{f_N = \frac{1}{2\delta t}}.$$

Change	Consequence
Smaller δt	Larger frequency range without aliasing
Longer DFT record T	Smaller frequency-bin spacing δf
Longer actual observation	Can improve frequency discrimination

Different quantities

$\delta t, \delta f$ describe sampling intervals; σ_t, σ_f describe signal energy spreads. Sampling imposes representation constraints, rather than a second intrinsic uncertainty principle.

Observation windows connect the two ideas

Observing through a finite window $w(t)$ gives

$$x_{\text{obs}}(t) = x(t)w(t) \quad \longleftrightarrow \quad X_{\text{obs}}(f) = (X * W)(f).$$

For a rectangular window of duration T_{obs} ,

$$W(f) = T_{\text{obs}} \operatorname{sinc}(fT_{\text{obs}}), \quad \operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}.$$

Its main-lobe width between the first zeros is $2/T_{\text{obs}}$. A short window broadens spectral features.

Zero-padding changes spacing, not resolving power

Appending zeros increases the DFT duration and decreases δf . It adds no observations and does not narrow the window's spectral main lobe.

Distinction: localization concerns the signal; sampling controls frequency range; observation duration helps determine frequency discrimination.